Research article

Journal of RAMAN SPECTROSCOPY

Received: 8 December 2016 Revised: 26 April 2017 Accepted: 3 May 2017 Published online in Wiley Online Library

(wileyonlinelibrary.com) DOI 10.1002/jrs.5177

Combined spectroscopic analysis of stratified glass eye beads from China dated to the Warring States Period

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We herein report the application of a combined spectroscopic approach for the analysis of eight stratified glass eye beads unearthed from China, which was dated back to the Warring States Period (475 BC to 221 BC). Portable X-ray fluorescence spectrometry (pXRF) was used to determine that seven samples were typical natron type soda–lime–silicate glass beads with low contents of MgO and K_2O except for one plant-ash type soda–lime–silicate sample. In addition, combined with micro-Raman spectroscopy, two different types of colorants were present in these beads, namely oxidized transition metals such as Co, Cu and Fe, and antimony-based compound opacifiers/colorants. Furthermore, the distribution of their glassy and scattering phases could be determined by the optical coherence tomography (OCT) images, in which it exhibits that their morphologies were related to their Sb contents. According to the determined typology, chemical composition, micro-crystal, and eyeball structural characteristics within the samples, the possible relations among them were speculated. It is the most interesting finding that seven natron type soda–lime glasses could be imported from the eastern Mediterranean area and one plant-ash type glass could be imported from the eastern region of the Euphrates River such as Iran area. Finally, through comparison with similar contemporary beads excavated from the famous tomb of Marquis Yi of the Zeng State (~433 BC) in the Hubei Province, provenance studies of these beads were carried out, and their origins and relationships are discussed briefly.
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Keywords: stratified glass eye beads; micro-Raman spectroscopy; pXRF; OCT; provenance study

Introduction

Ancient glass is an important artificial material for tracing mutual cultural and technical exchanges between the East and West.[1] For example, stratified glass eyebeads, which were popular during the Warring States Period in China, have been closely associated with social, economic and cultural development at that time.

The various samples selected for analysis have similar shapes and decorations to those found in the tomb of Marquis Yi of the Zeng State (~433 BC) in the Hubei Province. Over the past 20 years, a limited number of glass samples, and mainly fragments, have been analyzed from the tomb of Marquis Yi, and little Raman data have been published.[2–4] In 2014, we analyzed 46 intact representative samples from this tomb *in situ* in the Hubei Provincial Museum without any sample pretreatment. Although we obtained some valuable information about these beads, few details regarding the analysis of these and similar samples have been reported, due to the influence of weathering and limited contemporary reference data.

A number of studies have indicated that antimony-based opacifiers and colorants played an important role in the production of opaque glasses of the Mesopotamian, Egyptian and Roman Empire before the 2nd century BC.[5,6] In both Europe and the Byzantine Empire, it appears that antimonates were replaced by tin-based colorants and opacifiers for use in glass production by approximately the 4th century AD.[7] However, during the Renaissance Period, antimony-based opacifiers and colorants were employed extensively once again.[8] Tite therefore proposed a chronological order[7] for the use of antimony as an opacifying agent within the Mediterranean region. However, to the best of our knowledge, the majority of Chinese research has focused on ionic coloring agents and the chemical composition of ancient glasses, with few comparison studies being published in relation to the inorganic crystalline opacifiers and colorants used in the ancient glasses found in China.

In terms of potential analytical methods for studying these glasses, OCT is of particular interest. Indeed, it is one of the most effective tools in biomedicine,[9,10] materials science, film technology, industrial production[11] and archeological science,[12–15] due to its sensitivity, high spectral resolution and non-destructive nature. In the past few years, this technique has also been applied in the study of glazed cross-sectional structures and decorations.[16,17] However, it has not yet been widely reported for the characterization of ancient Chinese glasses. In addition to OCT, two alternative analytical techniques, namely micro-Raman spectroscopy[18–21] and pXRF, have been widely applied in the analysis of ancient artifacts, such as glasses,[22] glazes,[23] pigments and others.[24,25]

Thus, we report the analysis of stratified glass eye beads excavated from different locations in China and dating back to the Warring States Period (475 BC to 221 BC). We expect that the

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use of micro-Raman spectroscopy, pXRF and OCT will reveal details regarding the composition and preparation of these samples. Combined with the results obtained for beads excavated from the tomb of Marquis Yi and additional archeological research, the possible provenance of these beads is discussed. We expect that this research will shed new light on the glassmaking history, the routes of transmission and the native development of ancient Chinese glasses.

Materials and methods

Samples

A total of eight ancient glass eye beads dated to the Warring States Period were selected for analysis (Table S1). Five samples were provided by the Shanghai Museum of Glass and were intact. The final three samples were excavated as debris from the tomb of the Chu State in Xu-Jia-Ling, Xichuan County, Henan Province and the tomb of Marquis Yi of the Zeng State in Sui County, Hubei Province. The obtained beads were either opaque or translucent, and photographic images of the artifacts are shown in Fig. 1. In the context of surface weathering, all samples were initially analyzed *in situ* prior to sample pretreatment. Finally, the chemical compositions of different colors within these beads were measured for comparison.

Portable X-ray fluorescence spectrometry analysis

To gain a basic understanding of the chemical properties of the glass beads and the major elements present, eight stratified glass eye beads were subjected to pXRF (OURSTEX 100FA) prior to analysis by micro-Raman spectroscopy. A palladium (Pd) excitation tube was employed (40 or 15 kV, 0.5–1.0 mA) with an X-ray beam spot diameter on the sample surface of ~2.5 mm. In addition, a silicon drift detector with an active area of 5 mm^2 equipped with a polymer window (AP 3.3 film, Moxtek) was employed. The pXRF spectrometer allows the selection of direct, monochromatic and filter models to detect signals from light and heavy elements. All spectra were acquired over 300 s and were split into 100-s intervals for each model. In our previous study, this equipment exhibited good stability, small relative standard deviation values and good detection limits for the majority of elements of interest. Further details regarding the pXRF procedure can be found in the literature.[26,27]

Micro-Raman spectroscopy

The vibrational bands of the beads were then examined nondestructively by micro-Raman spectroscopy (LabRAM XploRA), using a 532-nm laser source, 1800 gr/mm diffraction grating, and 50× working distance objective. The laser was projected onto a focal spot (beam diameter, ~1 μm) of the artifact and the power maintained at <15 mW. The spectral resolution was <2 cm^{-1} over a spectral range of 100–4000 cm^{-1}. The detection system consisted of a multi-channel air-cooled charge coupled device, and the system was calibrated using a 520.7 cm^{-1} Si line from a single crystal Si wafer prior to each measurement.

Optical coherence tomography

The OCT system employed herein (HSL-2100, Santec) consisted of a swept light source with a central wavelength of 1315–1340 nm, a full scan wavelength range of 110 nm and a high scan speed of 20 kHz. The axial resolution of the OCT system was determined

Figure 1. Photographs of the analyzed stratified glass eye beads. [Colour figure can be viewed at wileyonlinelibrary.com]

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by the bandwidth of the swept source. As such, a 5.3-μm axial resolution in silicate materials with a 1.5 refractive index was estimated, along with a 12-μm lateral resolution in air. The imaging range was 0–20 mm. As indicated in the literature, this setup has already been employed in the analysis of porcelains.[16]

Results and discussion

Chemical composition of the stratified glass eye beads

Figure 2 shows the Raman spectra of all ancient glass eye beads' matrices. Their amorphous states can be discriminated by the Si—O stretching (i.e. ~1000 cm^{-1}) and bending (~500 cm^{-1}) envelopes. In order to focus on the microcrystals dispersed within these ancient glass eye beads, the typical features of glassy matrices are not discussed here in detail.

Table 1 outlines the chemical composition of the eight beads, providing the quantities of the major oxides present. As the X-ray beam spot diameter on the sample is about 2.5 mm, and the X-ray penetration depth is about 2–3 mm for natron type soda–lime–silicate glass, chemical compositions of white circles and dark blue eye balls may be somehow affected by adjacent areas. Comparison of natural and ground surfaces showed a significant loss in flux sodium, likely due to surface weathering. The weight percentage of Na_2O in the ground matrices ranged from 6.05 to 15.04%, with these specimens containing 7.38–8.40% CaO. In addition, literature data for the 44 intact samples excavated from the tomb of Marquis Yi reported Na_2O and CaO weight percentages of 1.69–6.44 and 7–13%, respectively.[28] Moreover, for the ground surface of bead sample HBWKI-18, which was excavated from the same tomb and exhibited a similar typology, the Na_2O content was in the range of 13.43–14.58%.[2] Thus, these beads could be classified as soda–lime silicate glass. With the exception of bead sample LMYZ02, all other stratified glass eye beads analyzed were characterized by low contents of MgO (<1.27%) and K_2O (<1.05%). We therefore deduced that seven glass beads in the Warring States Period could be categorized under the natron-type soda–lime silicate glass.[29] It should also be noted that sample LMYZ06's matrix and white circles both contained a higher PbO content (i.e. 6.82 and 5.31%) compared to the other beads examined herein, but there was a lower amount (i.e. 0.90%) in the cobalt blue eye ball in the centre. It is possible that high lead in the narrow white circles of LMYZ06 was detected due to its vicinity to the yellow substrate considering the probing area and X-ray penetration depth of pXRF. Meanwhile, as demonstrated in Table 1, higher MgO (1.71%) and K_2O (2.42%) contents indicate that the raw materials used to prepare sample LMYZ02 are different from those employed for the other beads. Thus, the bead could be classified as plant-ash type soda–lime silicate glass.

Upon consideration of their chemical compositions and typology characteristics, five samples (except for LMYZ02, LMYZ03 and LMYZ05) were similar to the beads from the tomb of Marquis Yi.[28] For example, XJL-Eyebead, HB-1 and HB-6 exhibited blue opaque matrices, white and ochre eye circles, and dark blue eyeballs. They generally contain little or no SnO_2, while the Sb_2O_3 content in their blue opaque matrix varies from 0.45 to 2.02%. In addition, LMYZ04 exhibits a blue–green translucent matrix and no ochre circle. For this sample, the SnO_2 contents of the matrices, white eye circles and dark blue eyeballs ranged from 0.04 to 0.21%, while the blue–green translucent matrix contained 1.31% Sb_2O_3. The bronze (slag) could have been used instead of copper. Furthermore, identification of the white SnO_2 was difficult, but this

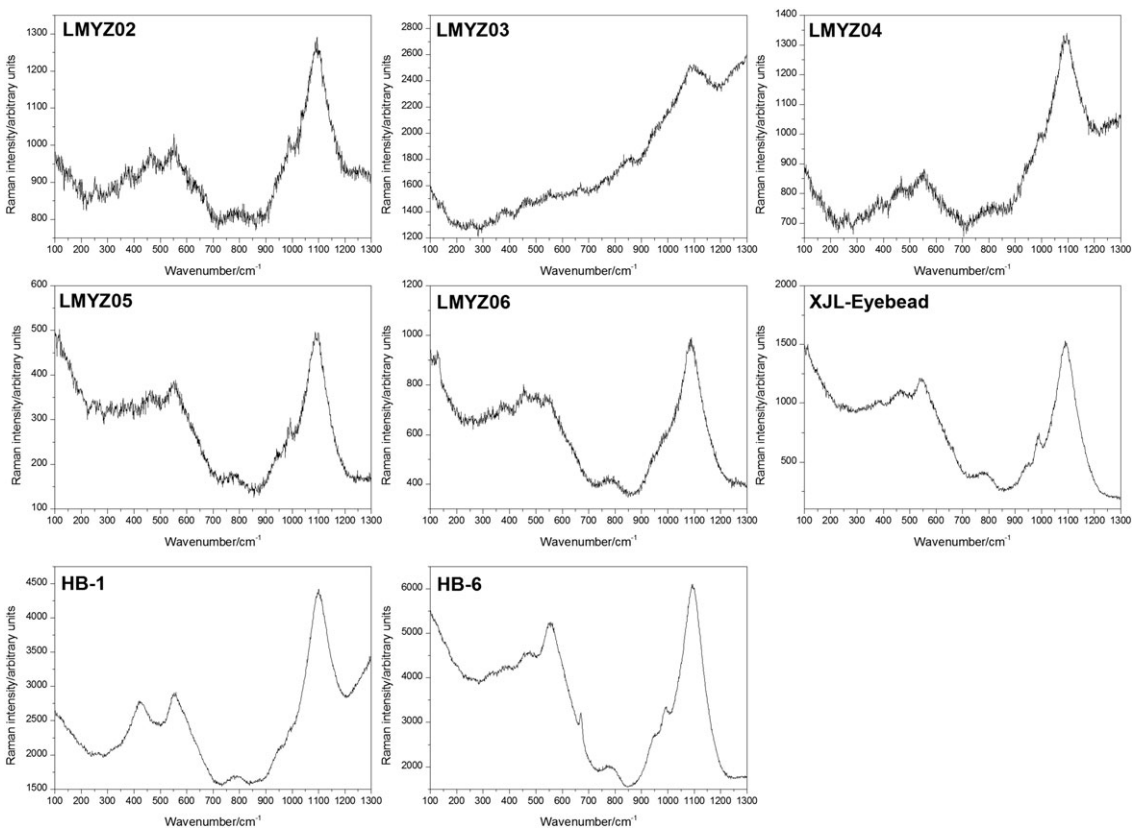

Figure 2. Raman spectra of all ancient glass eye beads' matrices.

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Table 1. Chemical composition determined by pXRF of the stratified glass eye beads from China (wt% oxides normalized to 100; n.d. = not detected)

Samples no.	Color of the measured point	Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	K_2O	CaO	MnO	CoO	Fe_2O_3	CuO	SnO_2	Sb_2O_3	PbO
LMYZ02	Blue matrix	5.74	1.79	3.85	73.19	0.40	2.54	8.12	0.29	0.45	1.84	0.26	n.d.	1.53	n.d.
	Ground blue matrix	6.05	1.71	3.54	73.39	0.13	2.42	7.80	0.21	0.34	1.58	0.20	n.d.	2.64	n.d.
	White circle	4.91	2.16	3.78	73.60	0.43	2.34	8.58	0.18	n.d.	1.06	0.19	n.d.	2.77	n.d.
	Dark blue eyeball	5.52	1.54	3.18	74.78	0.25	2.21	7.77	0.27	0.46	2.00	0.36	n.d.	1.66	n.d.
LMYZ03	Light green matrix	3.97	1.51	2.37	79.95	0.24	0.72	9.62	0.04	n.d.	1.17	0.11	n.d.	0.30	0.01
	Ground light green matrix	10.86	1.23	2.41	75.14	0.08	0.60	8.40	0.04	n.d.	0.89	0.10	n.d.	0.24	n.d.
	White circle	3.51	1.76	4.14	74.67	0.97	0.86	10.02	0.04	n.d.	1.19	0.04	n.d.	2.80	n.d.
	dark blue eyeball	5.00	1.38	3.81	73.89	0.96	0.81	9.37	0.03	0.43	1.84	0.30	n.d.	2.18	n.d.
LMYZ04	Blue–green matrix	4.17	1.53	6.68	75.62	0.01	0.57	6.99	0.11	n.d.	0.53	2.00	0.29	0.98	0.52
	Ground blue–green matrix	9.23	1.27	5.96	70.40	n.d.	0.64	8.13	0.09	n.d.	0.75	1.64	0.21	1.31	0.37
	White circle	4.41	1.40	7.33	71.10	0.15	0.33	10.20	0.09	n.d.	0.87	0.17	0.11	3.80	0.05
	Dark blue eyeball	6.58	1.26	5.48	69.92	0.25	0.46	10.12	0.05	0.45	1.92	0.49	0.04	2.93	0.06
LMYZ05	Blue matrix	7.93	1.04	2.57	72.79	0.25	0.71	8.55	0.09	0.51	1.63	1.17	0.08	2.66	0.02
	Ground blue matrix	10.08	1.11	4.39	70.44	0.02	1.05	7.43	0.07	0.74	2.23	1.84	0.09	0.48	0.02
	White circle	5.16	1.26	2.99	73.30	0.36	0.71	8.67	0.05	0.34	1.09	0.38	n.d.	5.67	0.02
	Dark blue eyeball	5.71	1.15	2.96	73.33	0.13	0.76	8.31	0.05	0.37	1.32	0.41	n.d.	5.48	0.02
LMYZ06	Yellow matrix	9.28	1.20	3.47	66.61	n.d.	0.59	9.96	0.03	n.d.	1.11	0.00	n.d.	0.84	6.91
	Ground yellow matrix	12.41	0.98	1.82	68.78	n.d.	0.23	7.38	0.02	n.d.	0.75	n.d.	n.d.	0.82	6.82
	White circle	7.99	1.10	3.49	66.57	n.d.	0.60	12.04	0.03	n.d.	0.97	0.00	n.d.	1.90	5.31
	Dark blue eyeball	7.88	1.05	0.00	80.45	n.d.	0.09	6.57	0.03	0.31	1.07	0.05	n.d.	1.58	0.90
XJL-Eyebead	Blue matrix	9.16	1.86	2.84	71.59	0.59	0.81	8.64	0.03	0.09	0.49	1.53	0.32	2.06	n.d.
	Ground blue matrix	13.44	1.24	2.64	71.23	0.20	0.91	7.42	0.02	n.d.	0.41	1.61	0.36	0.45	0.08
	White circle	5.88	1.46	2.45	70.84	n.d.	0.76	10.66	0.05	n.d.	0.55	n.d.	n.d.	7.36	n.d.
	Dark blue eyeball	9.34	1.60	2.40	68.95	0.32	0.45	9.87	0.04	0.26	1.66	0.24	n.d.	4.88	n.d.
HB-1[4]	Ground blue matrix	15.04	0.30	1.20	71.98	n.d.	0.30	8.24	n.d.	n.d.	0.32	1.75	n.d.	0.88	n.d.
	White circle	8.23	1.22	0.26	80.06	n.d.	0.12	7.40	0.03	n.d.	0.83	n.d.	n.d.	1.84	n.d.
HB-6[4]	Ground blue matrix	10.84	1.11	1.44	73.90	n.d.	0.37	8.30	0.04	n.d.	0.85	0.97	n.d.	2.02	0.18
	white circle	15.33	0.33	1.49	70.86	n.d.	0.37	8.42	0.03	n.d.	1.01	0.39	n.d.	1.12	0.65
	Dark blue eyeball	12.18	1.36	1.36	71.87	n.d.	0.32	7.80	0.03	0.32	1.69	0.21	n.d	2.32	0.54

was perhaps present as an impurity in Sb_2O_3. Interestingly, a stratified glass eye bead (Fig. S1) with similar decorations to LMYZ04 was excavated from the Fen-Shui-Ling tomb (early to middle of the Warring States Period) in Changzhi, Shanxi Province, but no chemical analysis has been reported.[30]Moreover, the LMYZ06 bead sample shows a yellow matrix which contained no traces of SnO_2, similar to the HBSBI-25 sample obtained from the tomb of Marquis Yi.

However, samples LMYZ02, LMYZ03 and LMYZ05 in terms of chemical composition, shape and decoration have not been previously observed in beads from this famous tomb. Samples LMYZ02 and LMYZ05 both exhibited dark blue translucent matrices and dark blue eyeballs. Both of these samples contained CoO (0.34–0.74 and 0.37–0.46%, respectively), while SnO_2 (0.09%) was only detected in the dark blue matrix of LMYZ05. In contrast, LMYZ03 exhibited a special color and decoration, and contained no traces of SnO_2. Indeed, 13 stratified glass eye beads (Fig. S2) with similar colors and decorations to LMYZ02 and LMYZ05 were unearthed from the Zhao Qing tomb (the Spring and Autumn Period) in Taiyuan, Shanxi Province.[31]

Identification of colorants and opacifiers

From examination of the stratified glass eye beads, it is clear that a range of colorants are employed in the different sections of the beads. Following analysis, we observed that for the matrices of XJL-Eyebead, HB-1 and HB-6, the average CuO content was 1.44%, while that of Fe_2O_3 was 0.53%. However, in LMYZ04, the contents of these two components were significantly higher, i.e. CuO = 1.64% and Fe_2O_3 = 0.75%. This resulted in blue–green matrix hues for LMYZ04 and blue hues for XJL-Eyebead, HB-1 and HB-6. We could therefore conclude that Cu^{2+} was the main colorant in the blue matrices of these samples. In contrast, LMYZ03 beads exhibited a light green matrix, in which the Fe_2O_3 content was higher than that of CuO (i.e. 0.89% and 0.10%).

In terms of the dark blue eyeballs of all samples, the Co, Fe and Mn contents ranged from 0.26 to 0.46%, 1.07 to 2% and 0.03 to 0.27%, respectively. Furthermore, the matrices of the LMYZ02 and LMYZ05 beads were also dark blue and contained CoO contents of 0.34 and 0.74%, respectively. In addition, compared with the dark blue eyeballs, their Fe_2O_3 contents (i.e. 1.58 and 2.23%) and MnO contents (i.e. 0.21 and 0.07%) were both comparable. Indeed, it is well known that cobalt ions are powerful chromophore agents that are intentionally designed and implemented by glass craftsmen during glass bead manufacture.

In contrast to the blue matrices and dark blue eyeballs, the white circles have lower contents of transition metals such as Cu and Fe, but have higher Sb_2O_3 contents (i.e. 1.12–7.36%). Table S2 shows the Raman signatures for the eight beads examined herein, where the white layers of all samples were characterized by their Raman peaks observed around 670 cm^{-1}. As shown in Fig. 3, the typical spectra of these samples are in good agreement with published values relating to the hexagonal $CaSb_2O_6$.[32] In addition, Fig. 4a shows the

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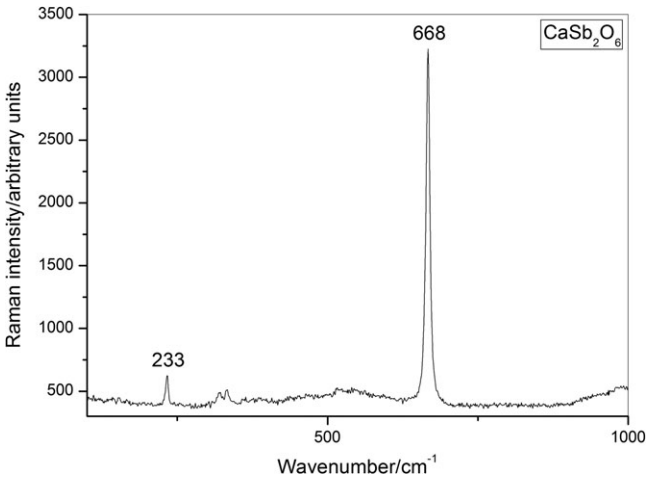

Figure 3. Raman spectra of the calcium antimonate ($CaSb_2O_6$) identified within XJL-Eyebead.

characteristic Raman bands at 478 and 631 cm^{-1} that coexist with $CaSb_2O_6$ in LMYZ06 and were assigned to the orthorhombic $Ca_2Sb_2O_7$.[33]

Moreover, an interesting phenomenon was observed in the LMYZ06 sample (Fig. 4b), where the spectrum of the yellow matrix reveals an intense scattering peak at 131 cm^{-1}, corresponding to the vibration of Pb_4O tetrahedra, or more specifically, to the lattice vibration of the Pb cation. In addition, the two overlapping modes at 329 cm^{-1} (E_g) and 297 cm^{-1} (F_{2g}) could be attributed to vibrational modes related to Sb—O and Pb—O bonds.[34] The strong A_{1g} mode assigned to the totally symmetric elongation of the SbO_6 octahedra was observed at 508 cm^{-1}, and the partial substitution of Sb^{5+} by Zn^{2+} was reflected by the presence of a shoulder at 455 cm^{-1}.[35] Thus, through comparison of these observations with literature data, we confirmed the presence of bindheimite $Pb_2Sb_2O_7$ in LMYZ06. Furthermore, the yellow color of the crystals induces the formation of yellow opaque glasses with colorless vitreous matrices. This result is consistent with the PbO (6.82%) and Sb_2O_3 (0.82%) contents determined by pXRF.

Additionally, very limited Pb was also found in the other samples. In view of the Mα peak of Pb and the Kα peak of S overlapping in pXRF, it may indicate that roasted lead or antimony sulfide was the raw material[35] used to prepare the lead antimonate yellow or calcium antimonate.

Cross-sectional structures of the eyeballs

The OCT images shown in Fig. 5 display the cross-sectional subsurface morphology of the samples based on their refractive indices or dielectric constants. Thus, the observed variation in image brightness, which is typically a gray scale variation from black to white, relates to the variation in local components and phase assemblage.[36]

As shown in Fig. 5, the OCT images exhibit various cross-sectional structures of six samples. Among the beads examined, the refractive indices were found to mutate in the presence of crystallized particles or unmelted materials in the samples' glassy phases. In addition, as shown, the penetration of each white circle was generally shallower than those of the dark blue eyeballs due to the higher antimony content in the white components enhancing the opacifying effect. Therefore, white pixels typically correspond to regions with high reflectivity or backscatter. In contrast, black pixels are displayed under weak backscattering conditions as there is no scattering in the homogeneous glassy phase. Thus, according to their respective structural characteristics, the distribution of glassy and scattering phases in these six beads could be distinguished easily from their OCT images.

As shown in Fig. 1, the various colors in samples are distributed in a range of patterns and layers. In addition, from Fig. 5, layer numbers for the various colors in the back of the dark blue eyeballs correspond to the circle numbers on the eyeball surfaces, and it results in the formation of a thin circular discoid eye that is thicker in the center. In general, the interfaces between differently colored layers are comparatively clear but tend to vary in thickness, decorative pattern, arc size and mutual position. We therefore expect that the layers of the dark blue eyeballs are constructed individually. Thus, from these observations, it appears

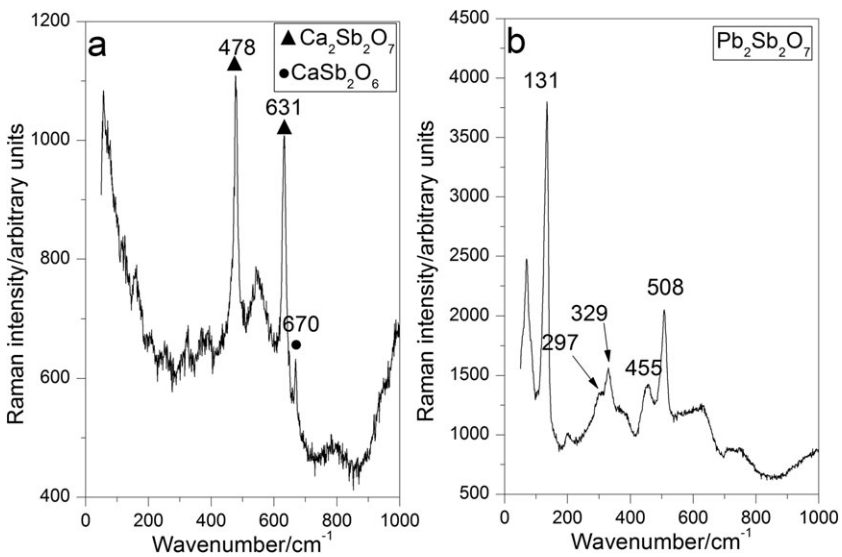

Figure 4. Raman spectra of the antimony-based opacifiers identified within LMYZ06: a, white circle, calcium antimonate ($CaSb_2O_6$, $Ca_2Sb_2O_7$); b, yellow matrix, bindheimite ($Pb_2Sb_2O_7$).

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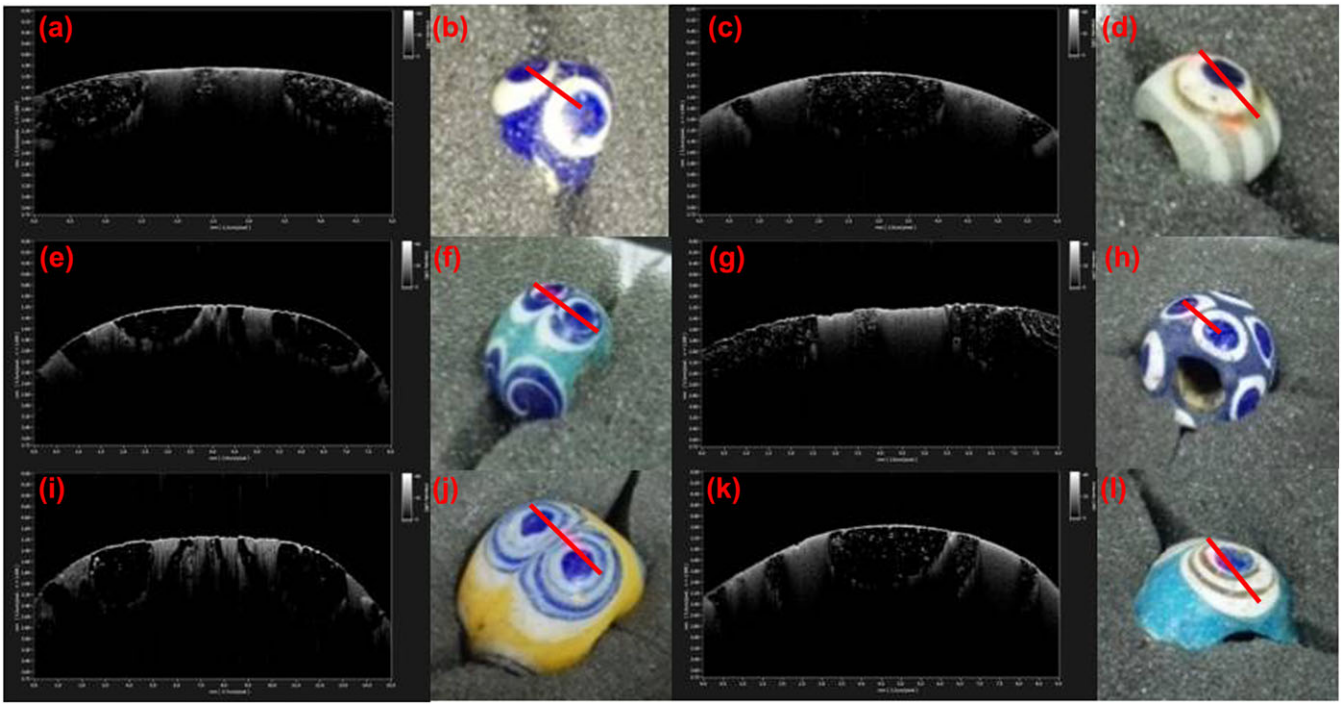

Figure 5. Oct images of six analyzed stratified glass eye beads. Test positions were indicated with red lines. (a) Oct images of LMYZ02; (b) test position of LMYZ02; (c) Oct images of LMYZ03; (d) test position of LMYZ03; (e) Oct images of LMYZ04; (f) test position of LMYZ04; (g) Oct images of LMYZ05; (h) test position of LMYZ05; (i) Oct images of LMYZ06; (j) test position of LMYZ06; (k) Oct images of XJL-Eyebead; (l) test position of XJL-Eyebead. [Colour figure can be viewed at wileyonlinelibrary.com]

that OCT images can provide important and direct evidence regarding the manufacture of stratified glass eye beads.

Remarks on the technology and provenance of stratified glass eye beads

Glass eye beads became popular worldwide following the reign of Amenhotep III (1391–1353 BC) in the late 18th Dynasty of the Egyptian New Kingdom. The technique for the manufacture of tabular eye beads composed of various glass layers was rediscovered in the 7th century BC, possibly in Egypt or in the eastern Mediterranean.[37] Indeed, a number of excellent examples of such early beads from the Mediterranean region (6th–2nd century BC), which are similar to the samples examined here, are also described in the literature.[38,39] According to published data, these beads are usually classified as soda–lime silicate glass. Recently, an ancient Egyptian necklace of stratified glass eye beads (Fig. 6), dating back 2300–2400 years ago, was excavated from the Altai Mountains in the central region of Siberia.[40] These beads were similar to some samples obtained from the tomb of Marquis

Figure 6. An ancient Egyptian necklace of stratified glass eye beads (dating back 2300–2400 years ago) from the Altai Mountains in the central region of Siberia. [Colour figure can be viewed at wileyonlinelibrary.com]

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Yi and three beads analyzed in this paper. However, the majorities of native glass eye beads found in China are dated from the middle/late Warring States Period to the Western Han dynasties (4th century BC to 1st century AD) and are composed of lead barium silicate glasses.[1]

We previously identified the use of antimony-based opacifiers and colorants in plant-ash type and natron type soda–lime silicate glass beads (1000–400 BC), which are distributed in the north of the Changjiang River.[4,35] Evidence from the Chinese glasses analyzed here support Tite's suggested chronology[7] for the use of antimony as an opacifying agent in the Mediterranean region. It may also indicate that mutual exchanges of ancient glass between China and the West were mainly through the desert and steppe routes of the Silk Road prior to the 4th century BC.[1] Based on the time sequence of the two hypotheses (*ex situ* and *in situ* crystallization) on antimony-based opacifiers,[6,41] we propose that the samples analyzed in our study were opacified by *ex situ* crystallization. Because $Ca_2Sb_2O_7$ crystals do not exist in nature, the 18th dynasty Egyptian craftsmen may have synthesized $Ca_2Sb_2O_7$ *ex situ* by using mixtures of Sb_2O_3 or Sb_2O_5 with calcium carbonates (atomic ratio Sb/Ca = 1) and treated mixtures between 1000 and 1100 °C.[41] Then, glasses are opacified by addition of these crystals $Ca_2Sb_2O_7$ through firing around 1000 °C for 1 h. To the best of our knowledge, from the 5th century BC to the 17th century AD antimony-based opacifiers were not used in native glasses (i.e. PbO–BaO–SiO_2, PbO–SiO_2 and K_2O–CaO–SiO_2) from central China.

In the context of cobalt glass colorants, the earliest example of their usage appears to be in a glass lump unearthed from Eridu in Iraq, which dated back to 2050 BC. During the reign of the Egyptian King Tuthmosis III (1479–1425 BC), cobalt colorants were extensively employed.[42,43] In addition, the use of CoO as a colorant for glass began in ancient China around the time of the Han dynasties (202 BC–220 AD).[3] The native cobalt ore found in China was known as asbolite[44] and contains a high percentage of MnO, as determined from the analysis of blue potash glass samples of the Eastern Han dynasty from Guangzhou.[45] However, in the case of the eye beads and dark blue eyeballs examined herein, CoO contents range from 0.26 to 0.46% and contain low manganese contents (0.03–0.27%) and high iron contents (1.07–2%). In addition, the CoO, MnO and Fe_2O_3 contents in the LMYZ02 and LMYZ05 matrices exhibited a similar trend to their eyeballs. These samples are similar to the contemporary glass eye beads unearthed from the tomb of Marquis Yi of the Zeng State (~433 century BC) in the Hubei Province[28] and from the Xu-Jia-Ling tomb[46] (5th–4th century BC) in the Henan Province. Thus, based on the analytical and archeological evidence, it appears that the cobalt ores present in the glass beads analyzed in this paper differ from historical Chinese samples.

In terms of the interior structural characteristics of the dark blue eyeball of the six samples studied by OCT and the archeological data on glass production technology in China at that time,[3] it can be speculated that these stratified glass eye beads were not manufactured by craftsmen of the Chu State.

In addition, we found that the contents of SnO_2 in seven natron type soda–lime silicate glass beads are various. Furthermore, samples LMYZ03, LMYZ05 and LMYZ06 exhibit characteristics similar to XJL-Eyebead, HB-1, HB-6 and LMYZ04 in the aspects of typology and chemical composition. Given these observations, the technologies employed to manufacture these beads appear to be diverse, likely originating from different glassmaking centers. The close relationships between these samples imply that mutual exchange exists between these centers.

Thus, through a careful combination of the evidence surrounding the typology, chemical composition, colorants, opacifiers and cross-sectional structures of the various beads of interest, we believe that seven stratified glass eye beads of natron type soda–lime silicate glass were imported from the eastern Mediterranean area, and the only one plant-ash type soda–lime silicate glass bead was imported from the eastern region of the Euphrates river such as Iran area. It also could be speculated that samples from Hubei and Henan Province (i.e. the tomb of Marquis Yi and from the Xu-Jia-Ling tomb) were introduced along the desert route of the Silk Road, while samples from Shanxi Province (i.e. the Fen-Shui-Ling tomb) were likely introduced along the steppe route of the Silk Road.

Conclusions

The complementary use of pXRF, micro-Raman spectrometry and OCT was employed for the analysis of eight stratified glass eye beads excavated from China. The material predominantly employed in the production of these glass beads was the natron-type soda–lime glass, and only one sample could be classified as plant-ash type soda–lime silicate glass with higher MgO and K_2O contents. We found that the transition metals in their oxidized state such as Cu and Fe ions affect the blue hues and the transparency of their matrices along with the different content of Sb. In addition, the presence of antimony-based opacifiers/colorants, such as $CaSb_2O_6$, $Ca_2Sb_2O_7$ and $Pb_2Sb_2O_7$ was determined based on the Raman spectra of the matrices. Furthermore, from the OCT images of six beads, the distribution of glassy and scattering phases could be determined, and their morphologies were found to be related to their Sb contents. Thus, based on the various decorations, chemical compositions, micro-crystals and eyeball structural characteristics within these beads, it was clear that the techniques employed to manufacture the different samples were diverse, and so the beads likely originated from different glassmaking centers with mutual exchanges in the eastern Mediterranean area and the eastern region of the Euphrates river such as Iran area.

Finally, in terms of the soda–lime glass eye beads dated to the Warring States Period and unearthed in China, we propose that the desert route of the Silk Road between the West and East prior to the 3rd century BC was the main trade route of glasses in the Hubei and Henan Provinces, while the steppe route of the Silk Road was likely the main route for glasses from the Shanxi Province. Future in-depth analysis of trace elements using laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) should be performed to provide further information regarding the origin of these stratified glass eye beads.

Acknowledgements

This research was supported by the National Natural Science Foundation of China (Grant No. 11374314 and Grant No. 51402326). The authors would like to thank the Institute of Cultural Relics in Henan province, the Hubei Provincial Museum and the Shanghai Museum of Glass for allowing the researchers to examine precious glass artifacts. We are also grateful to Dr. Song Liu and Curator Min Li for their valuable discussions during preparation of the manuscript.

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